

◆ SLIDE 1 — FRONT COVER

NETWORKING FUNDAMENTALS

Amazon VPC • Subnets • Internet Gateway

TEACHING NOTES

Welcome students into Week 3 of the bootcamp.

Explain that this week marks a major transition:
Students are now moving from:

✓ learning AWS services

to:

✓ understanding infrastructure architecture

Mention that networking is one of the foundations of:

- ◆ cloud security
- ◆ system communication
- ◆ infrastructure design
- ◆ architecture governance

Introduce the session focus:

✓ Amazon VPC

✓ Subnets

✓ Internet Gateway

Explain:

“Every modern cloud system depends on networking.”

Without networking:

- applications cannot communicate
- users cannot access systems
- cloud environments become insecure



◆ SLIDE 2 — SESSION OBJECTIVES

TEACHING NOTES

Walk students through the session goals slowly.

By the end of the session students should understand:

✓ what networking means in AWS

✓ why Amazon VPC exists

- ✓ public vs private subnet differences
- ✓ how Internet Gateways work
- ✓ how traffic flows inside AWS

Emphasize:

“The objective is not memorization.
The objective is architectural understanding.”

Tell students:

Networking affects:

- ✓ security
- ✓ communication
- ✓ performance
- ✓ exposure
- ✓ system stability

◆ SLIDE 3 — WHAT IS NETWORKING?

TEACHING NOTES

Define networking simply:

“Networking is how systems communicate.”

Examples:

- ✓ phones accessing websites
- ✓ applications reaching databases
- ✓ users opening cloud-hosted applications

Explain that in AWS, networking controls:

- ✓ connectivity
- ✓ communication paths
- ✓ internet access
- ✓ traffic flow
- ✓ isolation

Use this analogy:

“Networking is like roads connecting cities.”

Without roads:

- transportation stops

- communication stops

Without networking:

- applications stop functioning

Important clarification:

Networking is not only internet access.

It also controls:

- ✓ internal communication
- ✓ system segmentation
- ✓ security boundaries

◆ SLIDE 4 — WHY NETWORKING MATTERS

TEACHING NOTES

Explain that networking directly affects cloud security.

Poor networking can lead to:

- ✗ public database exposure
- ✗ unauthorized access
- ✗ cyberattacks
- ✗ unstable architecture

Mention that many cloud breaches happen because:

- ✓ ports were left open
- ✓ sensitive systems were public
- ✓ networking was poorly configured

Governance insight:

“Security begins during architecture design.”

Not after deployment.

Networking decisions are security decisions.

◆ SLIDE 5 — AMAZON VPC INTRODUCTION

TEACHING NOTES

Define Amazon VPC:

“Amazon VPC stands for Virtual Private Cloud.”

Explain:

A VPC allows organizations to create:

- ✓ isolated cloud networks
- ✓ controlled environments
- ✓ secure infrastructure boundaries

Use this analogy:

“A VPC is like your private secured section inside AWS.”

Inside the VPC organizations deploy:

- ✓ EC2 instances
- ✓ databases
- ✓ applications
- ✓ load balancers

Main purpose of a VPC:

- ✓ isolation
- ✓ control
- ✓ security



◆ SLIDE 6 — VPC COMPONENTS

TEACHING NOTES

Walk through each VPC component carefully.

Subnets

Smaller network sections inside the VPC.

Route Tables

Rules that control traffic flow.

Internet Gateway

Connects the VPC to the internet.

Security Groups

Virtual firewalls attached to instances.

NACLs

Subnet-level traffic filtering.

EC2 Instances

Compute resources running applications.

Important insight:

“A VPC is not just a network.
It is a complete controlled cloud environment.”

◆ SLIDE 7 — UNDERSTANDING SUBNETS

TEACHING NOTES

Define subnet simply:

“A subnet is a smaller section inside a VPC.”

Purpose of subnets:

- ✓ organization
- ✓ workload separation
- ✓ traffic control
- ✓ security segmentation

Use the building analogy:

“If the VPC is a building,
subnets are departments inside the building.”

Different departments require:

- ✓ different access levels
- ✓ different protections

Explain that not every system should be publicly accessible.

◆ SLIDE 8 — PUBLIC VS PRIVATE SUBNETS

TEACHING NOTES

Public Subnet

Explain:

Resources inside a public subnet can communicate with the internet.

Examples:

- ✓ web servers
- ✓ public applications
- ✓ load balancers

Private Subnet

Explain:

Private subnets protect sensitive systems from direct internet exposure.

Examples:

- ✓ databases
- ✓ internal applications
- ✓ backend systems



Critical teaching point:

“Sensitive systems should remain private whenever possible.”

Use architecture example:

System	Recommended Placement
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Website	Public subnet
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Database	Private subnet
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◆ SLIDE 9 — INTERNET GATEWAY

TEACHING NOTES

Define Internet Gateway (IGW):

“An Internet Gateway allows communication between a VPC and the internet.”

Without an IGW:

- ✗ websites cannot be reached
- ✗ EC2 instances cannot access internet resources

Use this analogy:

“The Internet Gateway is the main entrance connecting your cloud network to the outside world.”

Important clarification:

Even with an IGW:

- ✓ routing must still exist
- ✓ security rules still apply

◆ SLIDE 10 — HOW TRAFFIC FLOWS

TEACHING NOTES

Walk students through traffic flow step-by-step:

User → Internet → Internet Gateway → Public Subnet → EC2

Explain the process:

1. User opens a browser
2. Request travels through internet
3. AWS receives traffic
4. Internet Gateway allows entry
5. Public subnet routes request
6. EC2 responds to the user

Important insight:

“Cloud architecture depends heavily on controlled traffic flow.”

◆ SLIDE 11 — REAL-WORLD ARCHITECTURE SCENARIO

TEACHING NOTES

Use the online banking example.

Public Subnet

Hosts:

- ✓ website portal
- ✓ customer-facing applications

Private Subnet

Hosts:

- ✓ customer databases
- ✓ payment systems
- ✓ sensitive financial records

Ask students:

“Would you expose banking databases directly to the internet?”

Answer:

Absolutely not.

That is why:

- ✓ subnet separation matters
- ✓ architecture thinking matters

◆ SLIDE 12 — COMMON NETWORKING RISKS

TEACHING NOTES

Discuss common mistakes:

- ✗ public databases
- ✗ unrestricted SSH access
- ✗ open ports to the internet
- ✗ poor subnet segmentation

Explain possible consequences:

- ✓ breaches
- ✓ ransomware
- ✓ unauthorized access
- ✓ downtime

Governance insight:

“Misconfiguration is one of the biggest cloud security risks.”

◆ SLIDE 13 — NETWORKING BEST PRACTICES

TEACHING NOTES

Walk through each best practice.

- ✓ Use private subnets for sensitive systems
- ✓ Reduce unnecessary public exposure
- ✓ Separate workloads logically
- ✓ Monitor traffic
- ✓ Apply least privilege principles
- ✓ Document architecture properly

Enterprise insight:

Large organizations invest heavily in:

- ✓ network governance
- ✓ segmentation
- ✓ monitoring
- ✓ architecture security

Because networking affects every system.

◆ SLIDE 14 — GITHUB PROJECT REQUIREMENTS

TEACHING NOTES

Explain that GitHub work is:
NOT just homework.

It is:

- ✓ portfolio development
- ✓ proof-of-work
- ✓ engineering documentation

Students must create:

- ✓ vpc-basics.md
- ✓ subnet-design.md
- ✓ internet-gateway.md

Explain:

“Visible proof of learning creates professional credibility.”

Encourage students to:

- ✓ document clearly
- ✓ explain concepts properly
- ✓ upload screenshots professionally

◆ SLIDE 15 — KEY TAKEAWAYS

TEACHING NOTES

Reinforce the most important concepts:

- ✓ Amazon VPC creates isolated cloud networks
- ✓ Subnets organize infrastructure securely
- ✓ Public subnets allow internet communication
- ✓ Private subnets protect sensitive systems
- ✓ Internet Gateways enable internet access
- ✓ Networking directly affects cloud security

Most important lesson:

“If you cannot design the network correctly, you cannot secure the system correctly.”

◆ SLIDE 16 — FINAL CLOSING

TEACHING NOTES

Close the session strongly.

Tell students:

“Today you began transitioning from cloud users into cloud architects.”

Preview next session:

- ✓ Security Groups
- ✓ NACLs
- ✓ traffic filtering
- ✓ attack surface reduction

Final motivational close:

“The future belongs to engineers who understand systems deeply.”

Networking is one of the first major foundations of real cloud architecture thinking.

